

Homework #1

Answer the following questions. Write your answers to a document and send your answers with e-mail to dataorg2017@gmail.com until **October 30th, 2019 Wednesday**.

1. Consider a disk with a sector size 200 bytes, 400 tracks per surface, 100 sectors per track, 4 double-sided platters, and average seek time of 16 milliseconds.
 - a) What is the capacity of a platter?
 - b) What is the capacity of a track?
 - c) How many cylinders does the disk have?
 - d) What is the capacity of a cylinder?
 - e) If the disk platters rotate at 2500 rpm, what is the average rotational delay?
 - f) What is the block transfer time if one block contains 5 sectors?
2. Given the following $s = 16$ milliseconds, $r = 8.3$ milliseconds, $btt = 0.84$ milliseconds, $B = 2600$ bytes. A file having 80000 records and $R=400$ bytes is stored on this disk. Answer the following questions:
 - a) If this file is a pile file, calculate time to fetch a certain record.
 - b) If this file is a sorted sequential file with no overflow area, calculate time to fetch a certain record.
 - c) If this file is a sorted sequential file such that $1/4$ of the blocks are in the overflow area, calculate time to fetch a certain record.
 - d) If this file is an indexed file with a primary key index, calculate time to fetch a certain record given its primary key assuming that the whole primary key index can fit in main memory.
3. Assume that you try to estimate time required for batch update for a banking system. The master file has 100000 bank account information and each bank account record occupies 300 bytes. The transaction file contains 60000 transactions each occupies 300 bytes. Assume that no record is allowed to span more than one block, and 30% of the transactions are "open new account", 20% of the transactions are "close an account", and the others are "update" operations. The available memory for this operation is 3 buffer pages. Calculate a timing estimate for this batch update assuming that both the master and the transaction files are sorted with respect to the account number. Use $s = 16$ milliseconds, $r = 8.3$ milliseconds, $btt = 0.84$ milliseconds, $B = 2000$ bytes.